Alexa VanHattum

avh@cs.cornell.edu | cs.cornell.edu/~avh

Research Interests

Compilers, formal methods, programming languages

Education

Fall 2018–Present	**Cornell University** | Ph.D. Candidate, Computer Science
Fall 2021	**Cornell University** | M.S., Computer Science | GPA: 4.00
2012–2016	**Brown University** | B.Sc., Computer Science | GPA: 3.80

Publications

ASPLOS 2021	*Vectorization for Digital Signal Processors via Equality Saturation.* **Alexa VanHattum**, Rachit Nigam, Vincent T. Lee, James Bornholt, Adrian Sampson. PDF. ACM DL.
LCTES 2020	*A Synthesis-aided Compiler for DSP Architectures (WiP Paper).* **Alexa VanHattum***, Rachit Nigam*, Vincent T. Lee, James Bornholt, Adrian Sampson (*equal contribution). PDF. ACM DL.

Awards & Grants

April 2020	**NSF Graduate Research Fellow 2020**
April 2020	**Finalist: Qualcomm Innovation Fellowship 2020** with Rachit Nigam
May 2019	**Outstanding Teaching Assistant** from Cornell CS, for Programming Languages & Logics
April 2019	**Travel Scholarship** CRA-W Grad Cohort for Women Workshop
September 2018	**Travel Scholarship** Programming Languages Mentoring Workshop (PLMW) at the International Conference on Functional Programming (ICFP)
May 2016	**Honorable Mention: Senior Prize in CS** Brown CS
October 2015	**Anita Borg Scholar** Grace Hopper Celebration of Women in Computing

Research Positions

Fall 2018–Present	**Cornell University**, Graduate Research Advisor: Adrian Sampson Creating solver-aided compilers for heterogeneous hardware targets.
May–Dec 2021	**Amazon Web Services**, Automated Reasoning Applied Scientist Intern Mentor: Daniel Schwartz-Narbonne, Manager: Nathan Chong Improving dynamic dispatch for the Rust Model Checker.

Spring 2016	**Brown University**, Independent Study in Programming Languages Advisor: Shriram Krishnamurthi (Abstract) Used refinement types in Liquid Haskell to matrix arithmetic in R.
Summer 2014	**The University of Toledo**, Physics NSF REU Intern Advisor: Jaques G. Amar (Abstract) Implemented kinetic Monte Carlo simulations of particle island nucleation.

Industry Experience

2016–2018	**Apple**, Health Software Engineer Lead app/database engineer for Apple's first FDA-regulated irregular heart rhythm notification feature. Contributed to the HealthKit framework. Presented *What's New in Health* at Apple's 2017 WWDC conference.
Summer 2015	**Microsoft**, Software Engineering Intern Created a message search feature for the Windows Messaging Application.

Teaching Experience

Fall 2020	**Programming Languages & Logics**, Cornell University, Graduate TA
Fall 2018	**Programming Languages & Logics**, Cornell University, Graduate TA
Spring 2016	**Logic for Systems**, Brown University, Head TA
Fall 2017	**Intro. Object-Oriented Programming**, Brown University, Head TA
Spring 2015	**Discrete Structures & Probability**, Brown University, TA
Fall 2014	**Intro. Object-Oriented Programming**, Brown University, TA
Spring 2014	**First Byte of Computer Science**, Brown University, TA

Service & Outreach

Fall 2020–Present	**Programming Languages Discussion Group** Co-organizer
Summer 2020	**Artifact Evaluation Committee** SPLASH OOPSLA 2020
Summer-Fall 2020	**CS Teaching Committee** Student representative
Fall 2019–Present	**Graduate Students for Gender Inclusion in Computing** Secretary
Summer 2019	**Student Volunteer** PLDI 2019
Spring 2019	**Cornell CS PhD Visit Day**, Co-organizer, >100 visiting students
Spring 2019–Present	**Cornell CS Colloquium,** Organize student hosts for visiting speakers
Spring 2019	**Cornell Girls' Adventures in Math**, Student volunteer
2016–2017	**Apple Women in Science and Engineering Mentoring**, Mentor
2014–2016	**Brown Women in Computer Science**, Mentor
2012–2014	**Brown Algebra in Motion**, Site leader, math tutor at local high school